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Programme: - B.TECH (CSE)

Assignment: - Prepare the algorithm and test case table. Then write, compile, execute, and test the C programme against the prepared use cases. Write a menu-driven C programme using the switch-case to input a positive integer and perform one of the following operations based on the user's choice:

Count the number of even digits, and odd digits.

Find the sum of its even digits and the sum of its odd digits separately.

Part 1: - Thorough understanding of the problem statement

Input: - 1. A positive integer ('n').

2. An integer choice ('choice')

Output: - 1. If 'choice == 1': The total count of even digits and odd digits.

2. If 'choice == 2': The sum of all digits and the sum of all odd digits.

3. If 'choice' is invalid: An error message.

Formula/Logic : - 1. Extract the last digit: 'rem = n % 10'

2. Remove the last digit: 'n = n / 10'

3. Even check: 'if (rem % 2 == 0)'

4. Odd check: 'if (rem % 2 != 0)'

Part 2: - Writing, evaluating and evaluating the algorithm

Step 1: - Start

Step 2: - Declare integer variables: 'num', 'temp', 'choice', 'rem'.

Step 3: - Declare and initialize counters/accumulators: 'even_count = 0', 'odd_count = 0', 'even_sum = 0', 'odd_sum = 0'.

Step 4: - Prompt the user to input a positive integer and store it in 'num'.

Step 5: - Display the menu options:

1. Count even and odd digits.

2. Sum even and odd digits.

Step 6: - Prompt the user for 'choice'.

Step 7: - Store 'num' in 'temp' to avoid losing the original value during the loop.

Step 8: - Switch based on 'choice':

1. Case 1 (Count): 1. Loop while 'temp > 0':

2. 'rem = temp % 10'

3. If 'rem % 2 == 0', increment 'even_count' by 1.

4. Else, increment 'odd_count' by 1.

5. 'temp = temp / 10'

6. Print 'even_sum' and 'odd_sum'.

7. Break

2. Case 2 (Sum): 1. Loop while 'temp > 0' :

2. 'rem = temp % 10'

3. If 'rem % 2 == 0', add 'rem' to 'even_sum'.

4. Else, add 'rem' to 'odd_sum'.

5. 'temp = temp / 10'.

6. Print 'even_sum' and 'odd_sum'.

7. Break

3. Default: Print "Invalid choice!".

Step 9: - Stop

Part 3: - Writing and compiling the C programme

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File Machine View Input Devices Help

1 2 3 4 5

Session Actions Edit View Help

GNU nano 9.0

```
#include <stdio.h>

int main() {
    int num, temp, choice, rem;
    int even_count = 0, odd_count = 0;
    int even_sum = 0, odd_sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Error: Please enter a positive integer.\n");
        return 1;
    }

    printf("\n--- MENU ---\n");
    printf("1. Count the number of even and odd digits\n");
    printf("2. Find the sum of even and odd digits separately\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    temp = num;

    switch (choice) {
        case 1:
            // Handle edge case where input is exactly 0
            if (temp == 0) {
                even_count = 1;
            } else {
                while (temp > 0) {
                    rem = temp % 10;
                    if (rem % 2 == 0) {
                        even_count++;
                    } else {
                        odd_count++;
                    }
                    temp /= 10;
                }
            }
            printf("\nNumber: %d\n", num);
            printf("Even_dig = %d\n", even_count);
            printf("Odd_dig = %d\n", odd_count);
            break;

        case 2:
            while (temp > 0) {
                rem = temp % 10;
                if (rem % 2 == 0) {
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

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File Machine View Input Devices Help

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GNU nano 9.0

```
scanf("%d", &choice);

temp = num;

switch (choice) {
    case 1:
        // Handle edge case where input is exactly 0
        if (temp == 0) {
            even_count = 1;
        } else {
            while (temp > 0) {
                rem = temp % 10;
                if (rem % 2 == 0) {
                    even_count++;
                } else {
                    odd_count++;
                }
                temp /= 10;
            }
        }
        printf("\nNumber: %d\n", num);
        printf("Even_dlg = %d\n", even_count);
        printf("Odd_dlg = %d\n", odd_count);
        break;

    case 2:
        while (temp > 0) {
            rem = temp % 10;
            if (rem % 2 == 0) {
                even_sum += rem;
            } else {
                odd_sum += rem;
            }
            temp /= 10;
        }
        printf("\nNumber: %d\n", num);
        printf("Sum_even = %d\n", even_sum);
        printf("Sum_odd = %d\n", odd_sum);
        break;

    default:
        printf("\nInvalid choice! Please select 1 or 2.\n");
        break;
}

return 0;
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

Part 4: - Running and executing the algorithm on a platform (Kali Linux VM)

Step 1: - Open the terminal

Step 2: - Create the source code

Step 3: - Write and Save Code

Step 4: - Compile the programme

Step 5: - Run the Binary Executable programme

Session Actions Edit View Help

(kali@kali)-[~]

\$ nano digit_operations.c

(kali@kali)-[~]

\$ gcc digit_operations.c -o digit_operations

(kali@kali)-[~]

\$./digit_operations

Enter a positive integer: 12045

--- MENU ---

1. Count the number of even and odd digits
 2. Find the sum of even and odd digits separately
- Enter your choice (1 or 2): 1

Number: 12045

Even_dig = 3

Odd_dig = 2

(kali@kali)-[~]

\$./digit_operations

Enter a positive integer: 12345

--- MENU ---

1. Count the number of even and odd digits
 2. Find the sum of even and odd digits separately
- Enter your choice (1 or 2): 2

Number: 12345

Sum_even = 6

Sum_odd = 9

(kali@kali)-[~]

\$./digit_operations

Enter a positive integer: 2468

--- MENU ---

1. Count the number of even and odd digits
 2. Find the sum of even and odd digits separately
- Enter your choice (1 or 2): 1

Number: 2468

Even_dig = 4

Odd_dig = 0

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 2

Session Actions Edit View Help
Enter your choice (1 or 2): 1

Number: 2468
Even_dig = 4
Odd_dig = 0

(kali@kali)-[~]
$ ./digit_operations
Enter a positive integer: 1357

--- MENU ---
1. Count the number of even and odd digits
2. Find the sum of even and odd digits separately
Enter your choice (1 or 2): 2

Number: 1357
Sum_even = 0
Sum_odd = 16

(kali@kali)-[~]
$ ./digit_operations
Enter a positive integer: 0

--- MENU ---
1. Count the number of even and odd digits
2. Find the sum of even and odd digits separately
Enter your choice (1 or 2): 1

Number: 0
Even_dig = 1
Odd_dig = 0

(kali@kali)-[~]
$ ./digit_operations
Enter a positive integer: 9876

--- MENU ---
1. Count the number of even and odd digits
2. Find the sum of even and odd digits separately
Enter your choice (1 or 2): 3

Invalid choice! Please select 1 or 2.

(kali@kali)-[~]
```

```
(kali@kali)-[~]  
$ ./digit_operations  
Enter a positive integer: -50  
Error: Please enter a positive integer.
```

Part 5: - Compiling, executing and evaluating against all the test cases.

TEST CASE	INPUT NUMBER	INPUT CHOICE	EXPECTED CHOICE	ACTUAL OUTPUT	STATUS
TC1: Given Example 1	12045	1	Even_dig = 3 Odd_dig = 2	Even_dig = 3 Odd_dig = 2	PASS
TC2: Given Example 2	12345	2	Sum_even = 6 Sum_odd = 9	Sum_even = 6 Sum_odd = 9	PASS
TC3: All Even Digits	2468	1	Even_dig = 4 Odd_dig = 0	Even_dig = 4 Odd_dig = 0	PASS
TC4: All Odd Digits	1357	2	Sum_even = 0 Sum_odd = 16	Sum_even = 0 Sum_odd = 16	PASS
TC5: Single Zero	0	1	Even_dig = 1 Odd_dig = 0	Even_dig = 1 Odd_dig = 0	PASS
TC6: Invalid Menu Choice	9876	3	Invalid choice!	Invalid choice!	PASS
TC7: Negative Number	-50	1	Error: positive integer.	Error: positive integer.	PASS

Part 6: - Evaluation

The algorithm correctly addresses the problem by isolating the digit extraction logic into an efficient loop, where 'd' is the number of digits in the integer.